

ATTRIBUTE GRAMMAR FOR OFFSET FASE

AG = (G, A, R) where

A : {definition.offset , recordField.offset}

G:

(1) RecordType: type $\rightarrow$ recordfield*

(2) FuncDefinition : definition $\rightarrow$ type ID varDefinition* statement*

(3) VarDefinition: definition $\rightarrow$ type ID

(4) FunctionType : type $\rightarrow$ type varDefinition*

R:

Assuming localBytesSum and globalBytesSum are declared global variables

(1)

recordFieldBytesSum = 0

for(RecordField field : recordField*):

 field.offset = recordFieldBytesSum

 recordFieldBytesSum = recordFieldBytesSum + field.type.numberOfBytes()

(2)

localBytesSum = 0

(3)

if definition.scope == 0:

 definition.offset = globalBytesSum

 globalBytesSum = globalBytesSum + type.numberOfBytes()

else:

 localBytesSum = localBytesSum + type.numberOfBytes()

 definition.offset = -localBytesSum

(4)

paramBytesSum = 4

for (Vardefinition v: varDefinition*.reversed()):

 v.offset = paramBytesSum

 paramBytesSum += v.type.numberOfBytes()